Zenia **Zuraiq**

PH.D. STUDENT AND PRIME MINISTER'S RESEARCH FELLOW, INDIAN INSTITUTE OF SCIENCE

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Education

Indian Institute of Science (IISc) *Bengaluru, India*
PH.D. THEORETICAL PHYSICS [EXPECTED 2027] *2022 - present*

- Advisor: Prof. Banibrata Mukhopadhyay
- *Thesis*: Exploring unusual observations using massive, magnetized compact stars: theory and simulation

National Institute of Technology, Tiruchirappalli (NIT-T) *Tiruchirappalli, India*
M.SC. PHYSICS *2020-2022*

- Advisor: Dr. Somnath Mukhopadhyay
- *Thesis*: Effects of fermion mass and strong two-body interactions on the mass-radius relationship of non-rotating compact stars

Stella Maris College, University of Madras *Chennai, India*
B.SC. PHYSICS *2017-2020*

- *Dissertation*: Investigation on the optical properties of titanium nanoparticles in the treatment of waste water

Professional Experience

2022-present	**Prime Minister's Research Fellow (PMRF)**, Indian Institute of Science
2023-present	**Associate Member**, Indian Pulsar Timing Array (InPTA)

Publications

* - alphabetic ordering of authors

PUBLISHED

Major contributions

Zuraiq, Z., Das, M., Deb, D., Kalita, S., Weber, F., Mukhopadhyay, B. Anisotropic Compact Stars: Theory and Simulation from Microphysical Models to Macroscopic Structure and Observables. *Universe* 12 (**2026**) 5, 130.
DOI: https://doi.org/10.3390/universe12050130

Bhutani, A., Raj, N., **Zuraiq, Z.***. Dark, deep, deconfining: Phase transitions in neutron stars as powerful probes of hidden sectors. *Phys.Rev.D* 113 (**2026**) 10, 103046. DOI: https://doi.org/10.1103/yblk-411t, arXiv: 2507.08076

Zuraiq, Z., Mukhopadhyay, B., Weber, F. Massive neutron stars as mass gap candidates: Exploring equation of state and magnetic field. *Phys. Rev. D.* 109 (**2024**) 2, 023027.
DOI: https://doi.org/10.1103/PhysRevD.109.023027, arXiv: 2311.02169

Conference proceedings and white papers

Adhikari, P. *et al.* (61 authors including **Zuraiq, Z.**), Strongly interacting matter in extreme magnetic fields. *Prog. Part. Nucl. Phys.* 146 (**2026**) 104199. DOI: https://doi.org/10.1016/j.ppnp.2025.104199, arXiv: 2412.18632

Zuraiq, Z., Kumar, A., Hackett, A. J., Mukhopadhyay, B. Simulating super-Chandrasekhar white dwarfs. *World Scientific* (**2026**). DOI: https://doi.org/10.1142/14814 | *Proceedings of the Seventeenth Marcel Grossmann Meeting*

Zuraiq, Z. Kumar, A., Hackett, A. J., Bhattarai, S., Tout, C. A., Mukhopadhyay, B. Simulating super-Chandrasekhar white dwarfs. *Astrophys. Space Sci. Proc.* 61 (**2025**) 397-407.
DOI: https://doi.org/10.1007/978-3-031-90186-7_32, arXiv: 2411.18692 | *Book chapter*

Zuraiq, Z., Mukhopadhyay, B., Weber, F. Exploring massive neutron stars towards the mass gap: Constraining the high-density nuclear equation of state. *Astron. Rep.* 67 (**2023**) Suppl 2, S199-S206.
DOI: https://doi.org/10.1134/S1063772923140214 | *Proceedings of the Fifth Zeldovich Meeting*

As part of InPTA collaboration

Nobleson, K. *et al.* (36 authors including **Zuraiq, Z.**). The Indian Pulsar Timing Array Data Release 2: II. Customised single-pulsar noise analysis and noise budget. *JHEAp* 53 (**2026**) 100594.
DOI: https://doi.org/10.1016/j.jheap.2026.100594, arXiv: 2512.20455

Susobhanan, A. *et al.* (31 authors including **Zuraiq, Z.**). Revisiting wideband pulsar timing measurements. *Mon. Not. Roy. Astron. Soc.* 547 (**2026**) 4, stag444. DOI: https://doi.org/10.1093/mnras/stag444, arXiv: 2512.01288

Chowdhury, S. *et al.* (30 authors including **Zuraiq, Z.**). Effects of coronal mass ejection on PSR J1022+1001 and possible mode change of PSR J2145 - 0750 in the InPTA DR2. *JHEAp* 51 (**2026**) 100547.
DOI: https://doi.org/10.1016/j.jheap.2025.100547, arXiv: 2510.26594

Rana, P. *et al.* (47 authors including **Zuraiq, Z.**). The Indian Pulsar Timing Array Data Release 2: I. Dataset and Timing Analysis. *Publ. Astron. Soc. Aust.* 42 (**2025**) p.e108, DOI: https://doi.org/10.1017/pasa.2025.10066, arXiv: 2506.16769

Submitted

Zuraiq, Z., Mukhopadhyay, B., Kumar, A., Sarkar, A., Hackett, A. J., Banerjee, P., Tout, C. A. Evolution of magnetized main-sequence stars to super-Chandrasekhar white dwarfs by STARS simulation.

Zuraiq, Z., Mukhopadhyay, B. Anisotropic hybrid stars: Interplay of superconductivity and magnetic field leading to gravitational waves. arXiv: 2604.06308

Zuraiq, Z., Susobhanan, A. Improving Pulsar Timing Array sensitivity using VLBI Astrometry.

Mukhoapdhyay, B., Roy, N., **Zuraiq, Z.**, Chakraborty, C., Rao, A. R. Exploring PSR J0901-4046 as an ultra-slowly rotating neutron star versus a white dwarf.

Zuraiq, Z., Mukhopadhyay, B. Exploring the origin of slow-rotating pulsars: Emission from STARS simulated white dwarfs.

Tahbildar, H. *et al.* (31 authors including **Zuraiq, Z.**). The Indian Pulsar Timing Array Data Release 2: III. Search for a Stochastic Gravitational Wave Background. arXiv: 2608.02808

Chowdhury, S. *et al.* (29 authors including **Zuraiq, Z.**). Profile Reconstruction from Temporally Stable Emission Components for Timing PSR J1713+0747. arXiv: 2608.04108

Awards, Fellowships, & Grants

2022-2027	**Prime Minister's Research Fellowship**, Government of India
2022	**All India Rank 36 (out of 28135 candidates), awarded Junior Research Fellowship**, National Eligibility Test, Council of Scientific and Industrial Research (CSIR-NET), India
2022	**All India Rank 51 (out of 19375 candidates)**, Graduate Aptitude Test in Engineering (GATE)
2022	**All India Rank 198 (96.12 Percentile)**, Joint Entrance Screening Test (JEST)
2022	**Gold Medalist and Alumni Award for Outstanding Student**, M.Sc. Physics, NIT-T
2020	**All India Rank 669 (out of 16379 candidates)**, Joint Admission Test for Masters (JAM)
2020	**Silver Medalist and Outgoing student of the year**, B.Sc. Physics, Stella Maris College

Presentations

Talks

Anisotropic hybrid stars: Interplay of superconductivity and magnetic field leading to gravitational waves
6th Zeldovich Meeting, Pescara, Italy | 07/25

Simulating magnetized white dwarfs: Chandrasekhar limit and beyond
Workshop: Magnetism in low-mass stars and their compact remnants, IIT Bombay, Mumbai, India | 12/25

Anisotropic neutron stars as mass gap candidates
Brainstorming workshop on neutron stars, IMSc, Chennai, India | 08/25

Explaining unusual observations using magnetized white dwarfs
International Pulsar Timing Array (IPTA) Meeting 2025, California Institute of Technology, Pasadena, USA | 06/25

Addressing fundamental puzzles with the neutron star EOS
University of California, Riverside, USA | 06/25

Simulating magnetized white dwarfs: Chandrasekhar limit and beyond
GW Paleontology group meeting, University of California San Diego, USA | 06/25

Explaining unusual observations using magnetized white dwarfs
43rd Meeting of the Astronomical Society of India, NIT Rourkela, Odisha, India | 02/25

Simulating magnetized white dwarfs by time evolution: Chandrasekhar limit and beyond [**invited**]
4th Symposium of the BRICS Association on Gravity, Astrophysics and Cosmology, SGT University, Gurugram, India | 12/24

Anisotropic neutron stars as mass gap candidates
Conference on Classical and Quantum Gravity, CUSAT, Cochin, India | 11/24

A journey towards compact star physics... and beyond! [**invited**]
Research Day talk, Stella Maris College, Chennai, India | 09/24

Massive, magnetized neutron stars as mass gap objects
Parallel session: Galactic and extragalactic magnetars: recent observations and theoretical progress
17th Marcel Grossmann meeting, Pescara, Italy | 07/24

Simulating magnetized white dwarfs by time evolution: Chandrasekhar limit and beyond
Parallel session: Massive white dwarfs and related phenomena
17th Marcel Grossmann meeting, Pescara, Italy | 07/24

Simulating super-Chandrasekhar white dwarfs
International Symposium on Recent Development in Relativistic Astrophysics, SRM Sikkim, Gangtok, India | 12/23

Massive, magnetized compact stars: Theory and Simulation
10th International Conference on Gravitation and Cosmology, IIT Guwahati, Assam, India | 12/23

Massive neutron stars: Effect of EOS vs magnetic field [**invited**]
Online: London-Oldenburg Relativity Seminar | 10/23

Massive neutron stars: effects of EOSs and magnetic field
Online: Strongly interacting matter in extreme magnetic fields, ECT* workshop | 09/23

Exploring massive neutron stars towards the mass gap: Constraining the high density nuclear equation of state
5th Zeldovich Meeting, Yerevan, Armenia | 06/23

Posters

Simulating magnetized white dwarfs by time evolution: Chandrasekhar limit and beyond
24th European Workshop on White Dwarfs, ISTA Klosterneuburg, Austria | 08/26

Simulating magnetized white dwarfs by time evolution: Chandrasekhar limit and beyond
In-House Symposium 2025, Department of Physics, Indian Institute of Science, Bengaluru, India | 11/25

Explaining unusual observations using magnetized white dwarfs
In-House Symposium 2024, Department of Physics, Indian Institute of Science, Bengaluru, India | 11/24

Massive neutron stars as mass gap candidates [**invited**]
PMRF Annual Symposium 2024, IIT Indore, Indore, India | 03/24

Simulating super-Chandrasekhar white dwarf from main sequence star: Exploring stellar evolution codes STARS and MESA
Parallel session: Stars, Interstellar Medium, and Astrochemistry in Milky Way
42nd annual meeting of the Astronomical Society of India, Indian Institute of Science, Bengaluru, India | 02/24

Massive neutron stars as mass gap candidates [**Best poster award**]
In-House Symposium, Department of Physics, Indian Institute of Science, Bengaluru, India | 11/23

Simulating magnetised super-Chandrasekhar white dwarfs using STARS
3rd Conference on Plasma Simulations, Raman Science Center, IIA, Leh, India | 07/23

Conferences & other visits

Conferences attended

- *24th European Workshop on White Dwarfs*, ISTA Klosterneuburg, Austria | 08/26
- *6th Zeldovich Meeting*, Pescara, Italy | 07/26
- *Workshop: Magnetism in low-mass stars and their compact remnants*, IIT Bombay, Mumbai, India | 12/25
- *Brainstorming workshop on neutron stars*, IMSc, Chennai, India | 08/25
- *International Pulsar Timing Array (IPTA) Meeting 2025*, California Institute of Technology, Pasadena, USA | 06/25
- *43rd annual meeting of the Astronomical Society of India (ASI)*, NIT Rourkela, Rourkela, India | 02/25
- *4th Symposium of the BRICS Association on Gravity, Astrophysics and Cosmology,* SGT University, Gurgaon, India | 12/24
- *Conference on Classical and Quantum Gravity*, CUSAT, Cochin, India | 11/24
- *17th Marcel Grossmann meeting*, Pescara, Italy | 07/24
- *PMRF Annual Symposium 2024*, IIT Indore, Indore, India | 03/24
- *Meeting on Pulsar Timing Array Experiments: Present and Future of Indian Collaboration*, IMSc, Chennai, India | 02/24
- *42nd annual meeting of the Astronomical Society of India (ASI)*, Indian Institute of Science, Bengaluru, India | 02/24
- *International Symposium on Recent Developments in Relativistic Astrophysics*, SRM Sikkim, Gangtok, India | 12/23
- *10th International Conference on Gravitation and Cosmology (ICGC)*, IIT Guwahati, Guwahati, India | 12/23
- *3rd Conference on Plasma Simulations*, Raman Science Center, IIA, Leh, India | 07/23
- *5th Zeldovich meeting*, Yerevan, Armenia | 06/23

Research visits

- University of Warsaw | 07/26, 07/24
- ICRANet-Ferrara, University of Ferrara | 07/26, 06/24
- University of California, Riverside | 06/25
- University of California, San Diego | 06/25
- Department of Physics, San Diego State University | 06/25
- Instytut Fizyki Teoretycznej, Uniwersytet Wroclawski (University of Wroclaw) | 07/24

Internships and summer schools

- *IIA Summer Programme*, IIA, Bengaluru, India | 07/21
- *Vigyan Vidushi*, TIFR, Mumbai, India | 05-06/2021
- *Introductory Summer School in Astronomy and Astrophysics*, IUCAA, Pune, India | 05/21
- *National Institute on Undergraduate Sciences (NIUS) Camp (15.1)*, TIFR, Mumbai, India | 06/18
- *Summer Intern*, IIUCNN, Mahatma Gandhi University, Kottayam, India | 05/18

Teaching Experience

08/26 - present	**Mechanics and Waves II**, Teaching Assistant | Undergraduate course at Azim Premji University, Bengaluru, India
01-04/26	**Particle Physics and the Standard Model**, PMRF Teaching Assistant | NPTEL online course | Link
08-12/25	**Quantum Mechanics I**, Teaching Assistant | Undergraduate course at IISc
08-10/25	**Mechanics and Waves II**, Teaching Assistant | Undergraduate course at Azim Premji University, Bengaluru, India
07-10/25	**The Joy of Computing using Python**, PMRF Teaching Assistant | NPTEL online course | Link
01-04/25	**Intermediate Mechanics (Lab)**, Teaching Assistant | Undergraduate course at IISc
01-03/25	**Programming, Data Structures And Algorithms Using Python**, PMRF Teaching Assistant | NPTEL online course | Link
07-10/24	**The Joy of Computing using Python**, PMRF Teaching Assistant | NPTEL online course | Link
04-06/24	**Introductory Astronomy and Astrophysics**, Instructor, value added course at M.S. Ramaiah University of Applied Sciences, Bengaluru, India | Link
01-03/24	**Nuclear Astrophysics**, PMRF Teaching Assistant | NPTEL online course | | Link
08-12/23	**Introductory Mechanics**, Teaching Assistant | Undergraduate course at IISc [*Best TA award*]
08-12/23	**Introductory Astronomy**, Instructor to Classes VI-X, SPARC Public School, Bengaluru, India
01-06/23	**High school physics**, PMRF Teaching Assistant for Class XI, Kendriya Vidyalaya, IISc Bengaluru, India

Mentoring

2023	**Surendra Bhattarai**, Masters thesis student, IISER Kolkata, India [currently Ph.D. student at Yale University]
2023	**Ashmita Acharya**, Undergraduate student, M.S. Ramaiah University of Applied Sciences, Bengaluru, India [currently masters student at Uppsala University]

Skills

Programming Languages: Python, C++, MATLAB, Mathematica

Tools/Codes: LaTeX, MS Office, Cambridge STARS, MESA, Tempo2, ENTERPRISE

OS: Linux, Windows, macOS

Service & Outreach

Peer review: Reviewer for European Physical Journal C, Astrophysics and Space Science Proceedings

Public talk: *The Life and Death of Stars: Stellar Physics and Evolution*
IISc PRAVEGA Astrophysics Workshop, Bengaluru, India | 01/26, 05/26

Volunteer: IISc Open Day
Public Talk: *The Sun and our Solar System* (2023), Conducted quizzes (2024, 2026)

Public talk [**invited**]: *Looking out at the Universe in Radio Waves: From Communication to Astronomy*
IEEE ComSoc SCT SBC: ComConnect 1.0, SCT College of Engineering, Thiruvananthapuram, India | 09/25

Outreach talk: *Looking out at the universe: multi-wavelength astronomy*
SERB outreach program for class X students, IISc, Bengaluru, India | 10/24

Volunteer: 42nd annual meeting of the Astronomical Society of India, IISc, Bengaluru, India | 02/24

External expert: *National Workshop on physical sciences for CSIR-UGC NET and GATE aspirants*, NIT-T, India | 05/23

Outreach talks: *Overview of fermionic compact stars*
Talk I: Non-interacting stars and white dwarfs, Talk II: Introducing interactions
PMRF outreach talk series: Christ University, Bengaluru, India | 02-04/23

Talk: *Equation of motion of double pendulum using different mechanics and MATLAB*
NIT-T weekly colloquium | 06/21

Talk: *Exoplanet detection by transit method*
NIT-T weekly colloquium | 10/20

Public Talk: *Space Debris*
Loyola Astronomy and Rocketry Club, Chennai, India | 02/19